

NETWORK TRAFFIC ANALY

Wireshark PCAP Forensics Report

Submitted by: Tarun Thakur

University Roll no. : 2415001680

Course: Ethical Hacking — JOVAC

Assignment: Assignment 2 — PCAP Analysis Date:

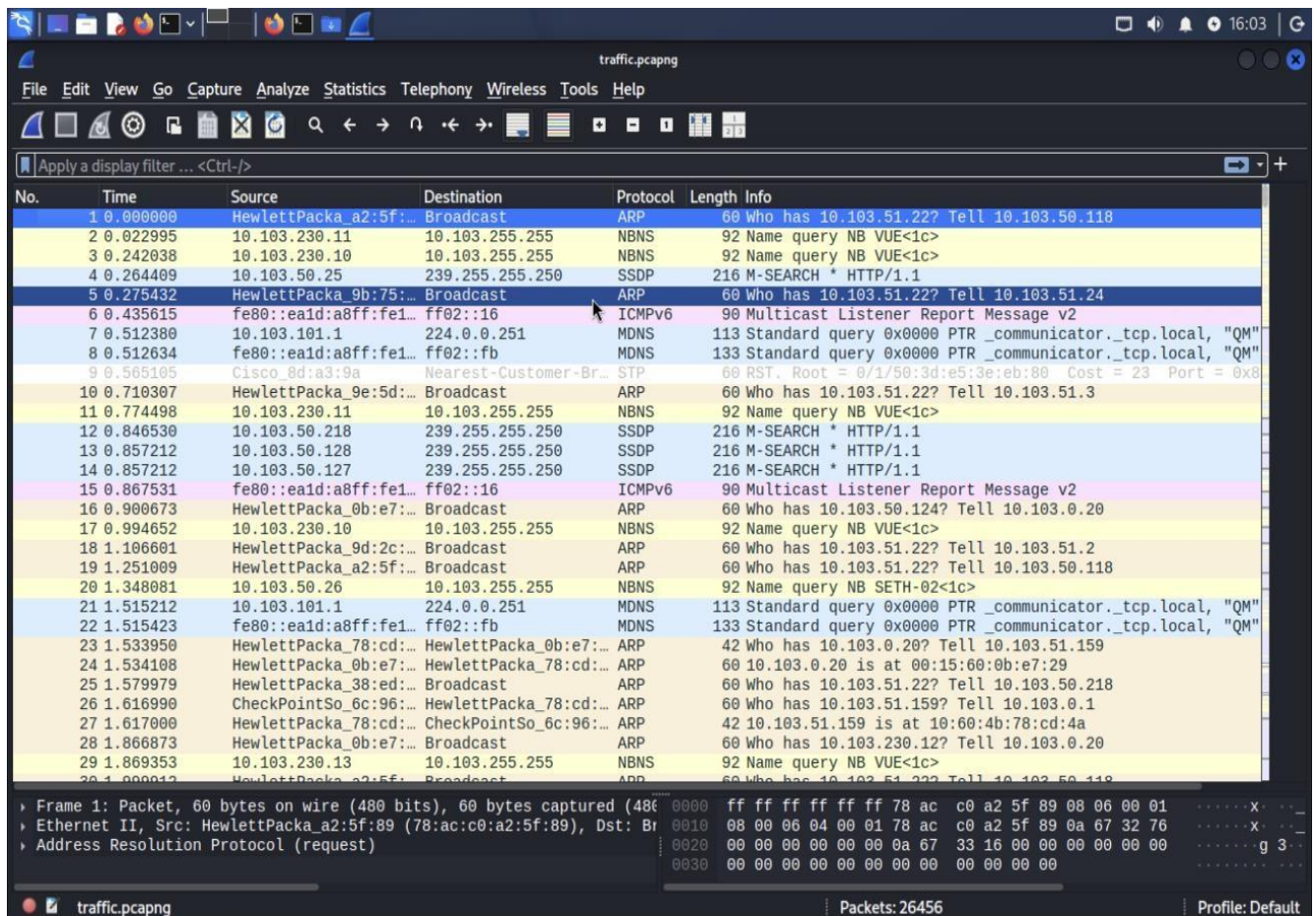
July 25, 2026

File Analyzed : traffic.pcapng

Tool : wireshark / Tshark 4.2.2

Objective : This report presents a detailed forensic analysis of the provided network packet capture file, **traffic.pcapng**, using **Wireshark** and **TShark**. The analysis focuses on extracting and verifying twenty key data points related to network communication, including protocol statistics, packet counts, IP and MAC address mappings, host communication patterns, vendor identification, and overall traffic characteristics. The primary objective of this assignment is to demonstrate practical skills in network traffic analysis, packet investigation, and the effective use of industry-standard network forensic tools, as required for the **JOVAC Ethical Hacking – Assignment 2**.

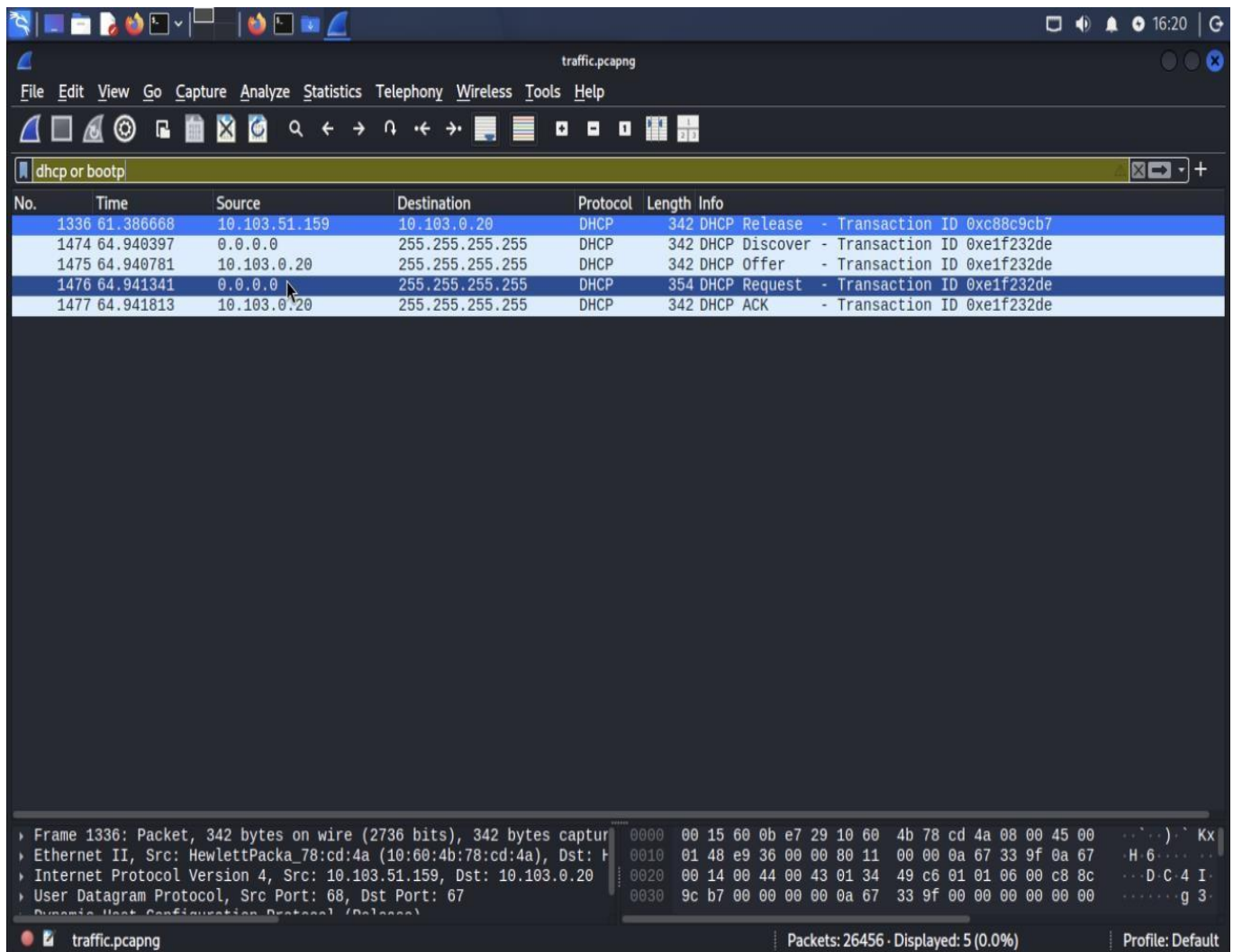
Capture File Overview : The following section presents a detailed analysis of each assignment question. Every answer has been obtained through Wireshark and TShark using appropriate display filters, packet-level inspection, and protocol analysis. All findings are based solely on the provided traffic.pcapng capture file to ensure accuracy and reproducibility



Detailed Question-by-Question Analysis :

E1. Download and analyze the network file; find the number of DHCP messages. (traffic (1))

Ans :- 5



The image shows a Wireshark network traffic capture window titled 'traffic.pcapng'. The filter bar at the top is set to 'dhcp or bootp'. The packet list pane displays five DHCP-related packets:

No.	Time	Source	Destination	Protocol	Length	Info
1336	61.386668	10.103.51.159	10.103.0.20	DHCP	342	DHCP Release - Transaction ID 0xc88c9cb7
1474	64.940397	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover - Transaction ID 0xe1f232de
1475	64.940781	10.103.0.20	255.255.255.255	DHCP	342	DHCP Offer - Transaction ID 0xe1f232de
1476	64.941341	0.0.0.0	255.255.255.255	DHCP	354	DHCP Request - Transaction ID 0xe1f232de
1477	64.941813	10.103.0.20	255.255.255.255	DHCP	342	DHCP ACK - Transaction ID 0xe1f232de

The packet details pane for the selected packet (Frame 1336) shows the following structure:

- Frame 1336: Packet, 342 bytes on wire (2736 bits), 342 bytes captured
- Ethernet II, Src: HewlettPacka_78:cd:4a (10:60:4b:78:cd:4a), Dst: 10:60:4b:78:cd:4a
- Internet Protocol Version 4, Src: 10.103.51.159, Dst: 10.103.0.20
- User Datagram Protocol, Src Port: 68, Dst Port: 67
- Dynamic Host Configuration Protocol (Release)

The packet bytes pane shows the raw data in hexadecimal and ASCII. The status bar at the bottom indicates 'Packets: 26456 · Displayed: 5 (0.0%)' and 'Profile: Default'.

Notes :- Breakdown: 1 DHCP Inform, 1 Discover, 1 Offer, 1 Request, 1 ACK

E2. Analyze the network file and find the number of ARP messages.

Ans :- 680 Packets

The screenshot displays the Wireshark network traffic analysis tool. The main window shows a list of captured packets, with the filter 'arp' applied. The list contains 26 packets, all of which are ARP messages. The selected packet (No. 1333) is an ARP request from Intel_6c:26:0a to Broadcast. The packet details pane shows the following structure:

- Frame 1333: Packet, 60 bytes on wire (480 bits), 60 bytes captured (480 bits) on interface 0
- Ethernet II, Src: HewlettPacka_9b:74:f7 (a0:48:1c:9b:74:f7), Dst: Broadcast (ff:ff:ff:ff:ff:ff)
- Address Resolution Protocol (request)

The packet bytes pane shows the raw data in hexadecimal and ASCII format:

```
0000 ff ff ff ff ff ff a0 48 1c 9b 74 f7 08 06 00 01 .....H..t
0010 00 00 06 04 00 01 a0 48 1c 9b 74 f7 0a 67 32 be .....H..t
0020 00 00 00 00 00 00 0a 67 33 16 00 00 00 00 00 00 .....g 3..
0030 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 .....

```

The status bar at the bottom indicates: Address Resolution Protocol (arp), 28 bytes. Packets: 26456 · Displayed: 680 (2.6%) Profile: Default.

E3. Analyze the network file and find the IP address that accessed 'baidu.com'.

Ans :- 10.103.51.159

The screenshot shows the Wireshark interface with a filter applied: `dns.qry.name contains "baidu"`. The packet list displays several DNS queries and responses. The selected packet is No. 20754, a Standard query from 10.103.51.159 to 10.103.0.20 for `www.baidu.com`. The packet details pane shows the structure of the DNS query, including Ethernet II, Internet Protocol Version 4, and User Datagram Protocol.

No.	Time	Source	Destination	Protocol	Length	Info
20754	303.328879	10.103.51.159	10.103.0.20	DNS	73	Standard query 0x8b7e A www.baidu.com
20755	303.331184	10.103.0.20	10.103.51.159	DNS	142	Standard query response 0x8b7e A www.baidu.com CNAME www
20823	304.130182	10.103.51.159	10.103.0.20	DNS	73	Standard query 0x1a40 A sp0.baidu.com
20825	304.201192	10.103.0.20	10.103.51.159	DNS	142	Standard query response 0x1a40 A sp0.baidu.com CNAME www
20826	304.203766	10.103.51.159	10.103.0.20	DNS	73	Standard query 0x5746 A sp1.baidu.com
20827	304.205616	10.103.0.20	10.103.51.159	DNS	142	Standard query response 0x5746 A sp1.baidu.com CNAME www
20828	304.207066	10.103.51.159	10.103.0.20	DNS	73	Standard query 0xe525 A sp2.baidu.com
20829	304.209201	10.103.0.20	10.103.51.159	DNS	142	Standard query response 0xe525 A sp2.baidu.com CNAME www
20894	304.880525	10.103.51.159	10.103.0.20	DNS	75	Standard query 0x36e3 A baike.baidu.com
20895	304.880525	10.103.51.159	10.103.0.20	DNS	73	Standard query 0xb9ec A b2b.baidu.com
20896	304.880553	10.103.51.159	10.103.0.20	DNS	76	Standard query 0x4157 A baobao.baidu.com
20897	304.884088	10.103.51.159	10.103.0.20	DNS	82	Standard query 0x171b A hectorstatic.baidu.com
20966	305.249350	10.103.0.20	10.103.51.159	DNS	116	Standard query response 0xb9ec A b2b.baidu.com CNAME b2b
20969	305.323014	10.103.0.20	10.103.51.159	DNS	122	Standard query response 0x4157 A baobao.baidu.com CNAME
20970	305.324575	10.103.51.159	10.103.0.20	DNS	76	Standard query 0x5973 A haokan.baidu.com
20971	305.330621	10.103.0.20	10.103.51.159	DNS	134	Standard query response 0x36e3 A baike.baidu.com CNAME b
20972	305.331623	10.103.51.159	10.103.0.20	DNS	74	Standard query 0x972e A help.baidu.com
20977	305.395209	10.103.0.20	10.103.51.159	DNS	250	Standard query response 0x972e A help.baidu.com A 110.24
20978	305.396834	10.103.51.159	10.103.0.20	DNS	74	Standard query 0xcea0 A home.baidu.com
20987	305.453558	10.103.0.20	10.103.51.159	DNS	122	Standard query response 0xcea0 A home.baidu.com A 183.23
20989	305.454811	10.103.51.159	10.103.0.20	DNS	75	Standard query 0x98fa A image.baidu.com
20993	305.520575	10.103.0.20	10.103.51.159	DNS	320	Standard query response 0x171b A hectorstatic.baidu.com
20997	305.534356	10.103.51.159	10.103.0.20	DNS	72	Standard query 0xa5aa A ir.baidu.com
21036	305.660962	10.103.0.20	10.103.51.159	DNS	162	Standard query response 0xa5aa A ir.baidu.com CNAME ir.b
21037	305.662353	10.103.51.159	10.103.0.20	DNS	77	Standard query 0xb46a A jingyan.baidu.com
21064	305.810768	10.103.0.20	10.103.51.159	DNS	151	Standard query response 0x5973 A haokan.baidu.com CNAME
21065	305.812277	10.103.51.159	10.103.0.20	DNS	74	Standard query 0xd2c0 A live.baidu.com
21066	305.868327	10.103.0.20	10.103.51.159	DNS	134	Standard query response 0xd2c0 A live.baidu.com CNAME po
21067	305.869886	10.103.51.159	10.103.0.20	DNS	73	Standard query 0x28a3 A map.baidu.com
21068	305.872010	10.103.0.20	10.103.51.159	DNS	150	Standard query response 0x09fa A image.baidu.com CNAME i

Frame 20754: Packet, 73 bytes on wire (584 bits), 73 bytes captured
Ethernet II, Src: HewlettPacka 78:cd:4a (10:60:4b:78:cd:4a), Dst: 10.103.0.20
Internet Protocol Version 4, Src: 10.103.51.159, Dst: 10.103.0.20
User Datagram Protocol, Src Port: 62733, Dst Port: 53

Packets: 26456 · Displayed: 86 (0.3%) Profile: Default

E4. Analyze the network file and find the number of packets the IP address 10.103.0.20 sent.

Ans :- 46G Packets

The screenshot shows the Wireshark network traffic analysis tool. The packet list pane displays a series of packets sent from the source IP 10.103.0.20 to the destination IP 10.103.51.159. The filter applied is `ip.src==10.103.0.20`. The selected packet (No. 20633) is a DNS Standard query response from `ssl.gstatic.com`. The packet details pane shows the structure of the packet: Ethernet II, Internet Protocol Version 4, User Datagram Protocol, and Domain Name System (response).

No.	Time	Source	Destination	Protocol	Length	Info
19633	257.284754	10.103.0.20	10.103.51.159	TCP	60	389 → 49869 [ACK] Seq=2830 Ack=520 Win=65017 Len=0
19643	258.014342	10.103.0.20	10.103.51.159	TCP	66	389 → 51983 [SYN, ACK] Seq=0 Ack=1 Win=16384 Len=0 MSS=1
19646	258.015403	10.103.0.20	10.103.51.159	LDAP	283	bindResponse(13255) saslBindInProgress , NTLMSSP_CHALLENGE
19648	258.016269	10.103.0.20	10.103.51.159	LDAP	79	bindResponse(13256) success
19650	258.016936	10.103.0.20	10.103.51.159	TCP	1514	389 → 51983 [ACK] Seq=255 Ack=315 Win=65221 Len=1460 [TC
19651	258.017025	10.103.0.20	10.103.51.159	LDAP	974	SASL GSS-API Privacy: payload (2360 bytes)
19654	258.017697	10.103.0.20	10.103.51.159	LDAP	248	SASL GSS-API Privacy: payload (174 bytes)
19657	258.018164	10.103.0.20	10.103.51.159	TCP	60	389 → 51983 [FIN, ACK] Seq=2829 Ack=519 Win=65017 Len=0
19659	258.018282	10.103.0.20	10.103.51.159	TCP	60	389 → 51983 [ACK] Seq=2830 Ack=520 Win=65017 Len=0
19798	269.741725	10.103.0.20	10.103.51.159	DNS	195	Standard query response 0xf52d A cdn.content.prod.cms.ms
19806	269.752103	10.103.0.20	10.103.51.159	DNS	196	Standard query response 0x9342 A assets.msn.com CNAME as
20266	296.129420	10.103.0.20	10.103.51.159	DNS	90	Standard query response 0xb234 A www.google.com A 142.25
20359	296.417799	10.103.0.20	10.103.51.159	DNS	91	Standard query response 0xa413 A www.gstatic.com A 142.2
20600	300.510425	10.103.0.20	10.103.51.159	DNS	111	Standard query response 0xa624 A ogs.google.com CNAME ww
20633	300.687754	10.103.0.20	10.103.51.159	DNS	91	Standard query response 0x1835 A ssl.gstatic.com A 142.2
20755	303.331184	10.103.0.20	10.103.51.159	DNS	142	Standard query response 0x8b7e A www.baidu.com CNAME ww
20825	304.201192	10.103.0.20	10.103.51.159	DNS	142	Standard query response 0x1a40 A sp0.baidu.com CNAME ww
20827	304.205616	10.103.0.20	10.103.51.159	DNS	142	Standard query response 0x5746 A sp1.baidu.com CNAME ww
20829	304.209201	10.103.0.20	10.103.51.159	DNS	142	Standard query response 0xe525 A sp2.baidu.com CNAME ww
20859	304.700089	10.103.0.20	10.103.51.159	DNS	157	Standard query response 0x438d A dss0.bdstatic.com CNAME
20860	304.737464	10.103.0.20	10.103.51.159	DNS	157	Standard query response 0x1f84 A dss1.bdstatic.com CNAME
20861	304.806446	10.103.0.20	10.103.51.159	DNS	160	Standard query response 0x1580 A ss1.bdstatic.com CNAME
20883	304.816636	10.103.0.20	10.103.51.159	DNS	103	Standard query response 0x30a3 A safebrowsing.googleapis
20966	305.249350	10.103.0.20	10.103.51.159	DNS	116	Standard query response 0xb9ec A b2b.baidu.com CNAME b2b
20969	305.323014	10.103.0.20	10.103.51.159	DNS	122	Standard query response 0x4157 A baobao.baidu.com CNAME
20971	305.330621	10.103.0.20	10.103.51.159	DNS	134	Standard query response 0x36e3 A baike.baidu.com CNAME b
20977	305.395209	10.103.0.20	10.103.51.159	DNS	250	Standard query response 0x972e A help.baidu.com A 110.24
20979	305.410198	10.103.0.20	10.103.51.159	DNS	156	Standard query response 0x5606 A ss.bding.com CNAME supe
20987	305.453558	10.103.0.20	10.103.51.159	DNS	122	Standard query response 0xcea0 A home.baidu.com A 183.23
20992	305.520575	10.103.0.20	10.103.51.159	DNS	220	Standard query response 0x174b A hectorstatic.baidu.com

Frame 20633: Packet, 91 bytes on wire (728 bits), 91 bytes captured
Ethernet II, Src: HewlettPacka_0b:e7:29 (00:15:00:0b:e7:29), Dst: 10.103.51.159
Internet Protocol Version 4, Src: 10.103.0.20, Dst: 10.103.51.159
User Datagram Protocol, Src Port: 53, Dst Port: 57203
Domain Name System (response)

Source Address: IPv4 address Packets: 26456 · Displayed: 469 (1.8%) Profile: Default

E5. Analyze the network file and find the number of UDP packets.

Ans:- G102 Packets

The screenshot shows the Wireshark interface with the 'udp' filter applied. The packet list displays various QUIC and DNS packets. Packet 20633 is selected, showing a DNS response from 10.103.0.20 to 10.103.51.159. The packet details pane shows the User Datagram Protocol section.

No.	Time	Source	Destination	Protocol	Length	Info
20619	300.678177	10.103.51.159	142.250.186.78	QUIC	75	Protected Payload (KP0), DCID=bb7b7321931eb5cf
20620	300.678262	142.250.186.78	10.103.51.159	QUIC	1392	Protected Payload (KP0)
20621	300.678326	10.103.51.159	142.250.186.78	QUIC	75	Protected Payload (KP0), DCID=bb7b7321931eb5cf
20622	300.678378	142.250.186.78	10.103.51.159	QUIC	1392	Protected Payload (KP0)
20623	300.679305	142.250.186.78	10.103.51.159	QUIC	1392	Protected Payload (KP0)
20624	300.679418	10.103.51.159	142.250.186.78	QUIC	75	Protected Payload (KP0), DCID=bb7b7321931eb5cf
20625	300.679418	142.250.186.78	10.103.51.159	QUIC	1392	Protected Payload (KP0)
20626	300.679506	142.250.186.78	10.103.51.159	QUIC	1392	Protected Payload (KP0)
20627	300.679565	10.103.51.159	142.250.186.78	QUIC	75	Protected Payload (KP0), DCID=bb7b7321931eb5cf
20628	300.680848	142.250.186.78	10.103.51.159	QUIC	1392	Protected Payload (KP0)
20629	300.680998	142.250.186.78	10.103.51.159	QUIC	1392	Protected Payload (KP0)
20630	300.681121	10.103.51.159	142.250.186.78	QUIC	75	Protected Payload (KP0), DCID=bb7b7321931eb5cf
20631	300.682534	142.250.186.78	10.103.51.159	QUIC	392	Protected Payload (KP0)
20632	300.687499	10.103.51.159	10.103.0.20	DNS	75	Standard query 0x1835 A ssl.gstatic.com
20633	300.687754	10.103.0.20	10.103.51.159	DNS	91	Standard query response 0x1835 A ssl.gstatic.com A 142.2
20634	300.689473	10.103.51.159	142.250.186.67	QUIC	1392	Initial, DCID=5bd88ac01024cf31, PKN: 1, CRYPTO, PADDING
20635	300.709113	10.103.51.159	142.250.186.78	QUIC	75	Protected Payload (KP0), DCID=bb7b7321931eb5cf
20636	300.757450	142.250.186.67	10.103.51.159	QUIC	1392	Protected Payload (KP0)
20637	300.758152	10.103.51.159	142.250.186.67	QUIC	206	Protected Payload (KP0), DCID=5bd88ac01024cf31
20638	300.758532	142.250.186.78	10.103.51.159	QUIC	68	Protected Payload (KP0)
20639	300.773609	10.103.51.159	142.250.184.228	QUIC	315	Protected Payload (KP0), DCID=4a4a93f775a6a8db
20640	300.815802	10.103.51.159	142.250.185.110	QUIC	1392	Initial, DCID=1c386a935bdae3cc, PKN: 1, CRYPTO, PADDING
20641	300.817591	142.250.186.67	10.103.51.159	QUIC	664	Protected Payload (KP0)
20642	300.817591	142.250.186.67	10.103.51.159	QUIC	119	Protected Payload (KP0)
20643	300.817862	10.103.51.159	142.250.186.67	QUIC	75	Protected Payload (KP0), DCID=5bd88ac01024cf31
20644	300.829112	142.250.184.228	10.103.51.159	QUIC	71	Protected Payload (KP0)
20645	300.842882	142.250.186.67	10.103.51.159	QUIC	67	Protected Payload (KP0)
20646	300.844963	10.103.51.159	142.250.184.228	QUIC	75	Protected Payload (KP0), DCID=4a4a93f775a6a8db
20647	300.873811	10.103.51.101	239.255.255.250	SSDP	216	M-SEARCH * HTTP/1.1
20648	300.879874	142.250.185.110	10.103.51.159	QUIC	1392	Protected Payload (KP0)

Frame 20633: Packet, 91 bytes on wire (728 bits), 91 bytes captured
Ethernet II, Src: HewlettPacka_0b:e7:29 (00:15:00:0b:e7:29), Dst: 10.103.51.159
Internet Protocol Version 4, Src: 10.103.0.20, Dst: 10.103.51.159
User Datagram Protocol, Src Port: 53, Dst Port: 57203
Domain Name System (response)

User Datagram Protocol: Protocol

Packets: 26456 · Displayed: 9102 (34.4%) Profile: Default

E6. Analyze the network file and find the number of SMB packets.

Ans :- 518 Packets

The screenshot displays the Wireshark network traffic analysis tool. The main window shows a list of captured packets, with the filter 'smb or smb2' applied. The packet list is sorted by time, and the selected packet (No. 20595) is highlighted in blue. The packet details pane on the right shows the structure of the selected packet, including the Ethernet II header, Internet Protocol Version 4 header, and User Datagram Protocol header. The packet is identified as an SMB2 (Server Message Block Protocol version 2) packet.

No.	Time	Source	Destination	Protocol	Length	Info
20139	287.477087	10.103.50.245	10.103.255.255	BROWSER	243	Host Announcement 409_04, Workstation, Server, SQL Serve
20177	288.561175	10.103.51.100	10.103.255.255	SMB_NE...	280	SAM LOGON request from client[Malformed Packet]
20193	289.212617	10.103.50.80	10.103.255.255	SMB_NE...	280	SAM LOGON request from client[Malformed Packet]
20230	291.547787	10.103.50.82	10.103.255.255	SMB_NE...	280	SAM LOGON request from client[Malformed Packet]
20233	292.024916	10.103.50.208	10.103.255.255	SMB_NE...	280	SAM LOGON request from client[Malformed Packet]
20240	293.145026	10.103.51.19	10.103.255.255	SMB_NE...	282	SAM LOGON request from client[Malformed Packet]
20250	293.975914	10.103.51.138	10.103.255.255	SMB_NE...	280	SAM LOGON request from client[Malformed Packet]
20254	294.485368	10.103.50.83	10.103.255.255	SMB_NE...	280	SAM LOGON request from client[Malformed Packet]
20263	295.997346	10.103.51.141	10.103.255.255	BROWSER	243	Host Announcement 409_08, Workstation, Server, SQL Serve
20398	296.783177	10.103.50.194	10.103.255.255	SMB_NE...	280	SAM LOGON request from client[Malformed Packet]
20408	297.277775	10.103.51.138	10.103.255.255	BROWSER	243	Host Announcement 409_01, Workstation, Server, SQL Serve
20527	298.746841	10.103.50.86	10.103.255.255	SMB_NE...	280	SAM LOGON request from client[Malformed Packet]
20589	299.707839	10.103.50.5	10.103.255.255	BROWSER	243	Host Announcement LAPTOP-4P07CS8S, Workstation, Server,
20592	299.929401	10.103.50.184	10.103.255.255	SMB_NE...	280	SAM LOGON request from client[Malformed Packet]
20595	300.156789	10.103.50.163	10.103.255.255	SMB_NE...	280	SAM LOGON request from client[Malformed Packet]
20744	302.686797	10.103.50.129	10.103.255.255	SMB_NE...	282	SAM LOGON request from client[Malformed Packet]
20747	303.043967	10.103.51.142	10.103.255.255	SMB_NE...	280	SAM LOGON request from client[Malformed Packet]
20749	303.193974	10.103.51.107	10.103.255.255	SMB_NE...	280	SAM LOGON request from client[Malformed Packet]
20752	303.256316	10.103.50.85	10.103.255.255	SMB_NE...	280	SAM LOGON request from client[Malformed Packet]
20759	303.376889	10.103.50.41	10.103.255.255	SMB_NE...	280	SAM LOGON request from client[Malformed Packet]
20764	303.666957	10.103.51.109	10.103.255.255	SMB_NE...	280	SAM LOGON request from client[Malformed Packet]
20771	303.866085	10.103.51.141	10.103.255.255	SMB_NE...	280	SAM LOGON request from client[Malformed Packet]
20915	305.105681	10.103.50.39	10.103.255.255	SMB_NE...	280	SAM LOGON request from client[Malformed Packet]
21062	305.784435	10.103.51.1	10.103.255.255	SMB_NE...	282	SAM LOGON request from client[Malformed Packet]
21190	306.478651	10.103.51.20	10.103.255.255	SMB_NE...	282	SAM LOGON request from client[Malformed Packet]
21192	306.498811	10.103.51.39	10.103.255.255	SMB_NE...	282	SAM LOGON request from client[Malformed Packet]
21217	306.885479	10.103.50.233	10.103.255.255	SMB_NE...	282	SAM LOGON request from client[Malformed Packet]
21227	307.072882	10.103.50.8	10.103.255.255	SMB_NE...	282	SAM LOGON request from client[Malformed Packet]
21234	307.353686	10.103.50.37	10.103.255.255	SMB_NE...	280	SAM LOGON request from client[Malformed Packet]
21236	307.370404	10.103.51.12	10.103.255.255	SMB_NE...	282	SAM LOGON request from client[Malformed Packet]

Frame 20595: Packet, 280 bytes on wire (2240 bits), 280 bytes captured on interface 0 (eth0) from 10.103.50.163 to 10.103.255.255, Ethernet II, Src: Hewlett-Packard (88:3d:26:88:3d:26), Dst: 01:00:5e:00:00:00, Internet Protocol Version 4, Src: 10.103.50.163, Dst: 10.103.255.255, User Datagram Protocol, Src Port: 138, Dst Port: 138

SMB2 (Server Message Block Protocol version 2): Protocol

Packets: 26456 - Displayed: 518 (2.0%)

Profile: Default

E7. Analyze the network file and find the number of packets sent to the source IP address 10.103.0.20.

Ans :- 547 Packets (ip.dst==10.103.0.20)

The image shows a Wireshark network traffic analysis interface. The filter bar at the top displays the filter `ip.dst==10.103.0.20`. The packet list pane shows a list of packets, with the following columns: No., Time, Source, Destination, Protocol, and Length. The packets are filtered to show only those sent to the destination IP address 10.103.0.20. The packet details pane shows the details of the selected packet (No. 20358), including Ethernet II, Internet Protocol Version 4, and User Datagram Protocol. The packet bytes pane shows the raw data of the selected packet.

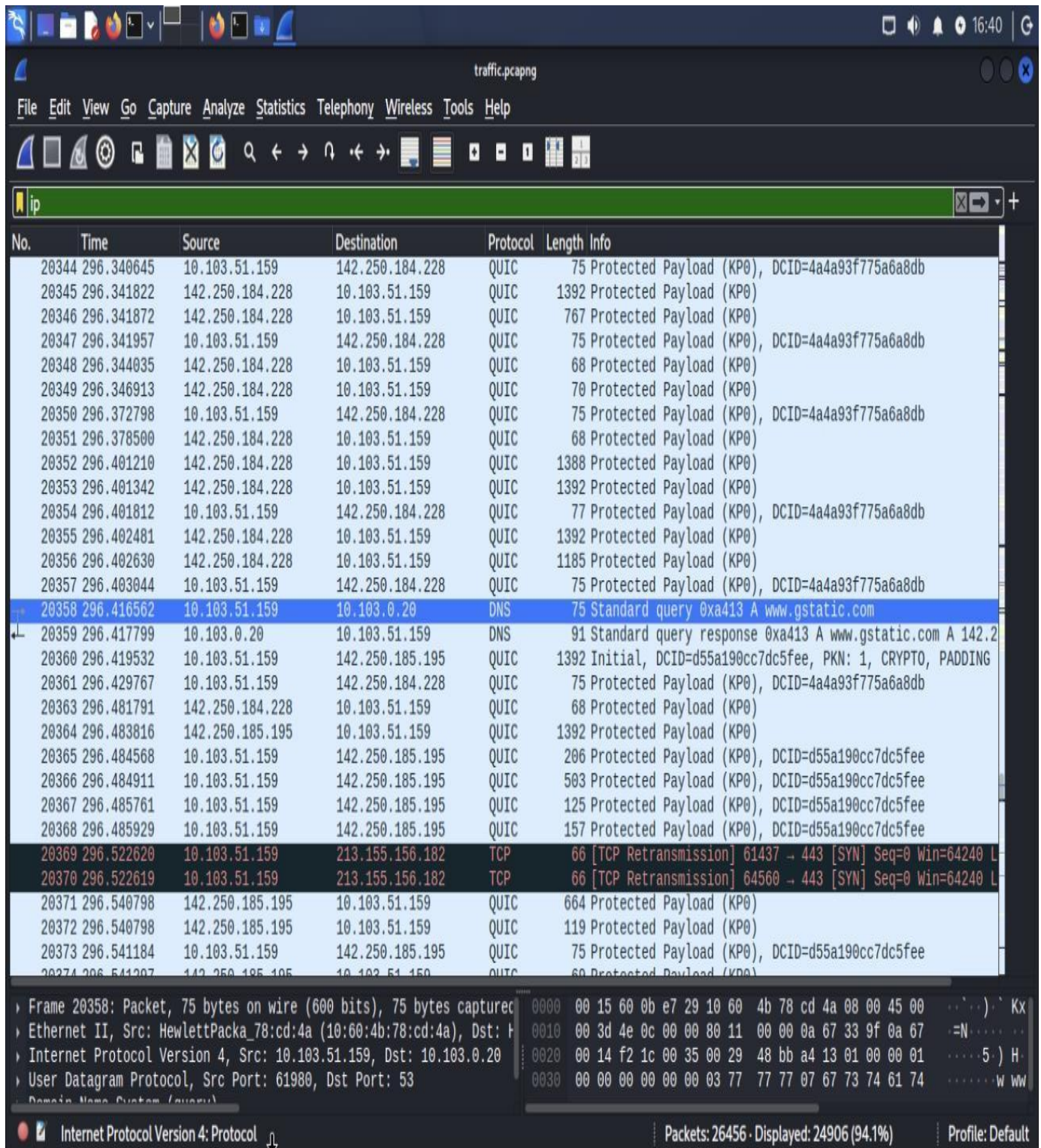
No.	Time	Source	Destination	Protocol	Length	Info
19632	257.284676	10.103.51.159	10.103.0.20	TCP	54	49869 → 389 [ACK] Seq=520 Ack=2830 Win=2102016 Len=0
19642	258.014179	10.103.51.159	10.103.0.20	TCP	66	51983 → 389 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 WS=256
19644	258.014408	10.103.51.159	10.103.0.20	TCP	54	51983 → 389 [ACK] Seq=1 Ack=1 Win=2102272 Len=0
19645	258.015047	10.103.51.159	10.103.0.20	LDAP	135	bindRequest(13255) "<ROOT>", NTLMSSP_NEGOTIATEsasl
19647	258.015725	10.103.51.159	10.103.0.20	LDAP	214	bindRequest(13256) "<ROOT>", NTLMSSP_AUTH, User: \sas\
19649	258.016384	10.103.51.159	10.103.0.20	LDAP	127	SASL GSS-API Privacy: payload (53 bytes)
19652	258.017046	10.103.51.159	10.103.0.20	TCP	54	51983 → 389 [ACK] Seq=315 Ack=2635 Win=2102272 Len=0
19653	258.017417	10.103.51.159	10.103.0.20	LDAP	226	SASL GSS-API Privacy: payload (152 bytes)
19655	258.017938	10.103.51.159	10.103.0.20	LDAP	86	SASL GSS-API Privacy: payload (12 bytes)
19656	258.018086	10.103.51.159	10.103.0.20	TCP	54	51983 → 389 [FIN, ACK] Seq=519 Ack=2829 Win=2102016 Len=0
19658	258.018197	10.103.51.159	10.103.0.20	TCP	54	51983 → 389 [ACK] Seq=520 Ack=2830 Win=2102016 Len=0
19797	269.741416	10.103.51.159	10.103.0.20	DNS	88	Standard query 0xf52d A cdn.content.prod.cms.msn.com
19805	269.751901	10.103.51.159	10.103.0.20	DNS	74	Standard query 0x9342 A assets.msn.com
20265	296.129207	10.103.51.159	10.103.0.20	DNS	74	Standard query 0xb234 A www.google.com
20358	296.416562	10.103.51.159	10.103.0.20	DNS	75	Standard query 0xa413 A www.gstatic.com
20599	300.508422	10.103.51.159	10.103.0.20	DNS	74	Standard query 0xa624 A ogs.google.com
20632	300.687499	10.103.51.159	10.103.0.20	DNS	75	Standard query 0x1835 A ssl.gstatic.com
20754	303.328879	10.103.51.159	10.103.0.20	DNS	73	Standard query 0x8b7e A www.baidu.com
20821	304.130181	10.103.51.159	10.103.0.20	DNS	77	Standard query 0x438d A dss0.bdstatic.com
20822	304.130182	10.103.51.159	10.103.0.20	DNS	77	Standard query 0x1f84 A dss1.bdstatic.com
20823	304.130182	10.103.51.159	10.103.0.20	DNS	73	Standard query 0x1a40 A sp0.baidu.com
20826	304.203766	10.103.51.159	10.103.0.20	DNS	73	Standard query 0x5746 A sp1.baidu.com
20828	304.207066	10.103.51.159	10.103.0.20	DNS	73	Standard query 0xe525 A sp2.baidu.com
20830	304.210110	10.103.51.159	10.103.0.20	DNS	76	Standard query 0x1580 A ss1.bdstatic.com
20881	304.813911	10.103.51.159	10.103.0.20	DNS	72	Standard query 0x5606 A ss.bdimg.com
20882	304.816438	10.103.51.159	10.103.0.20	DNS	87	Standard query 0x30a3 A safebrowsing.googleapis.com
20894	304.880525	10.103.51.159	10.103.0.20	DNS	75	Standard query 0x36e3 A baike.baidu.com
20895	304.880525	10.103.51.159	10.103.0.20	DNS	73	Standard query 0xb9ec A b2b.baidu.com
20896	304.880553	10.103.51.159	10.103.0.20	DNS	76	Standard query 0x4157 A baobao.baidu.com
20897	304.884089	10.103.51.159	10.103.0.20	DNS	82	Standard query 0x173b A baobaostatic.baidu.com

Frame 20358: Packet, 75 bytes on wire (600 bits), 75 bytes captured
Ethernet II, Src: HewlettPacka_78:cd:4a (10:60:4b:78:cd:4a), Dst: 10.103.0.20
Internet Protocol Version 4, Src: 10.103.51.159, Dst: 10.103.0.20
User Datagram Protocol, Src Port: 61980, Dst Port: 53

Packets: 26456 · Displayed: 547 (2.1%) Profile: Default

E8. Examine the network file and isolate only the IPv4 traffic; determine the count of packets shown.

Ans :- 24G06 Packets



The image shows a Wireshark network traffic analysis interface. The top menu bar includes File, Edit, View, Go, Capture, Analyze, Statistics, Telephony, Wireless, Tools, and Help. Below the menu is a toolbar with various icons for packet capture and analysis. The main display area shows a list of captured packets, with the 'ip' filter applied. The selected packet (20358) is highlighted in blue. The packet details pane on the right shows the selected packet's structure: Ethernet II, Internet Protocol Version 4, and User Datagram Protocol. The status bar at the bottom indicates that 26456 packets were captured and 24906 were displayed (94.1%).

No.	Time	Source	Destination	Protocol	Length	Info
20344	296.340645	10.103.51.159	142.250.184.228	QUIC	75	Protected Payload (KP0), DCID=4a4a93f775a6a8db
20345	296.341822	142.250.184.228	10.103.51.159	QUIC	1392	Protected Payload (KP0)
20346	296.341872	142.250.184.228	10.103.51.159	QUIC	767	Protected Payload (KP0)
20347	296.341957	10.103.51.159	142.250.184.228	QUIC	75	Protected Payload (KP0), DCID=4a4a93f775a6a8db
20348	296.344035	142.250.184.228	10.103.51.159	QUIC	68	Protected Payload (KP0)
20349	296.346913	142.250.184.228	10.103.51.159	QUIC	70	Protected Payload (KP0)
20350	296.372798	10.103.51.159	142.250.184.228	QUIC	75	Protected Payload (KP0), DCID=4a4a93f775a6a8db
20351	296.378500	142.250.184.228	10.103.51.159	QUIC	68	Protected Payload (KP0)
20352	296.401210	142.250.184.228	10.103.51.159	QUIC	1388	Protected Payload (KP0)
20353	296.401342	142.250.184.228	10.103.51.159	QUIC	1392	Protected Payload (KP0)
20354	296.401812	10.103.51.159	142.250.184.228	QUIC	77	Protected Payload (KP0), DCID=4a4a93f775a6a8db
20355	296.402481	142.250.184.228	10.103.51.159	QUIC	1392	Protected Payload (KP0)
20356	296.402630	142.250.184.228	10.103.51.159	QUIC	1185	Protected Payload (KP0)
20357	296.403044	10.103.51.159	142.250.184.228	QUIC	75	Protected Payload (KP0), DCID=4a4a93f775a6a8db
20358	296.416562	10.103.51.159	10.103.0.20	DNS	75	Standard query 0xa413 A www.gstatic.com
20359	296.417799	10.103.0.20	10.103.51.159	DNS	91	Standard query response 0xa413 A www.gstatic.com A 142.2
20360	296.419532	10.103.51.159	142.250.185.195	QUIC	1392	Initial, DCID=d55a190cc7dc5fee, PKN: 1, CRYPTO, PADDING
20361	296.429767	10.103.51.159	142.250.184.228	QUIC	75	Protected Payload (KP0), DCID=4a4a93f775a6a8db
20363	296.481791	142.250.184.228	10.103.51.159	QUIC	68	Protected Payload (KP0)
20364	296.483816	142.250.185.195	10.103.51.159	QUIC	1392	Protected Payload (KP0)
20365	296.484568	10.103.51.159	142.250.185.195	QUIC	206	Protected Payload (KP0), DCID=d55a190cc7dc5fee
20366	296.484911	10.103.51.159	142.250.185.195	QUIC	503	Protected Payload (KP0), DCID=d55a190cc7dc5fee
20367	296.485761	10.103.51.159	142.250.185.195	QUIC	125	Protected Payload (KP0), DCID=d55a190cc7dc5fee
20368	296.485929	10.103.51.159	142.250.185.195	QUIC	157	Protected Payload (KP0), DCID=d55a190cc7dc5fee
20369	296.522620	10.103.51.159	213.155.156.182	TCP	66	[TCP Retransmission] 61437 → 443 [SYN] Seq=0 Win=64240 L
20370	296.522619	10.103.51.159	213.155.156.182	TCP	66	[TCP Retransmission] 64560 → 443 [SYN] Seq=0 Win=64240 L
20371	296.540798	142.250.185.195	10.103.51.159	QUIC	664	Protected Payload (KP0)
20372	296.540798	142.250.185.195	10.103.51.159	QUIC	119	Protected Payload (KP0)
20373	296.541184	10.103.51.159	142.250.185.195	QUIC	75	Protected Payload (KP0), DCID=d55a190cc7dc5fee
20374	296.541207	142.250.185.195	10.103.51.159	QUIC	60	Protected Payload (KP0)

Frame 20358: Packet, 75 bytes on wire (600 bits), 75 bytes captured
Ethernet II, Src: HewlettPacka_78:cd:4a (10:60:4b:78:cd:4a), Dst: 10:103:0:20
Internet Protocol Version 4, Src: 10.103.51.159, Dst: 10.103.0.20
User Datagram Protocol, Src Port: 61980, Dst Port: 53
Domain Name System (query)

Internet Protocol Version 4: Protocol

Packets: 26456 · Displayed: 24906 (94.1%)

Profile: Default

EG. Analyze the network file and examine the last SMB packet; identify the source IP address.

Ans :- Source IP = 10.103.50.202

The screenshot displays the Wireshark interface with a packet capture named 'traffic.pcapng'. The filter bar shows 'smb or smb2'. The packet list pane contains 30 entries, with the last entry (No. 26452) selected. The details pane shows the structure of the selected packet: Ethernet II, Internet Protocol Version 4, User Datagram Protocol, and NetBIOS Datagram Service.

No.	Time	Source	Destination	Protocol	Length	Info
23499	326.108830	10.103.51.149	10.103.255.255	SMB_NE...	280	SAM LOGON request from client[Malformed Packet]
23550	327.026571	10.103.50.81	10.103.255.255	SMB_NE...	280	SAM LOGON request from client[Malformed Packet]
25160	329.283275	10.103.51.150	10.103.255.255	BROWSER	243	Host Announcement 409_13, Workstation, Server, SQL Serve
25198	329.505032	10.103.50.214	10.103.255.255	SMB_NE...	280	SAM LOGON request from client[Malformed Packet]
25201	329.551542	10.103.51.96	10.103.255.255	SMB_NE...	280	SAM LOGON request from client[Malformed Packet]
25217	330.200007	10.103.51.144	10.103.255.255	SMB_NE...	280	SAM LOGON request from client[Malformed Packet]
25229	331.062627	10.103.51.29	10.103.255.255	BROWSER	243	Host Announcement 409_12, Workstation, Server, SQL Serve
25259	331.425749	10.103.50.173	10.103.255.255	SMB_NE...	280	SAM LOGON request from client[Malformed Packet]
25261	331.488641	10.103.51.101	10.103.255.255	SMB_NE...	280	SAM LOGON request from client[Malformed Packet]
25266	331.639899	10.103.50.72	10.103.255.255	SMB_NE...	280	SAM LOGON request from client[Malformed Packet]
25353	336.669558	10.103.51.132	10.103.255.255	SMB_NE...	282	SAM LOGON request from client[Malformed Packet]
26070	337.828621	10.103.51.152	10.103.255.255	BROWSER	243	Host Announcement 409_18, Workstation, Server, SQL Serve
26206	338.406746	10.103.51.151	10.103.255.255	BROWSER	243	Host Announcement 409_19, Workstation, Server, SQL Serve
26285	339.125395	10.103.51.154	10.103.255.255	SMB_NE...	280	SAM LOGON request from client[Malformed Packet]
26291	339.277822	10.103.50.40	10.103.255.255	SMB_NE...	280	SAM LOGON request from client[Malformed Packet]
26313	339.883735	10.103.50.175	10.103.255.255	BROWSER	243	Host Announcement SETB-03, Workstation, Server, NT Works
26338	341.144851	10.103.50.188	10.103.255.255	SMB_NE...	280	SAM LOGON request from client[Malformed Packet]
26341	341.441479	10.103.50.183	10.103.255.255	SMB_NE...	280	SAM LOGON request from client[Malformed Packet]
26343	342.201530	10.103.51.92	10.103.255.255	SMB_NE...	280	SAM LOGON request from client[Malformed Packet]
26351	342.790849	10.103.50.213	10.103.255.255	SMB_NE...	280	SAM LOGON request from client[Malformed Packet]
26353	342.881173	10.103.50.166	10.103.255.255	SMB_NE...	280	SAM LOGON request from client[Malformed Packet]
26364	344.203370	10.103.50.43	10.103.255.255	SMB_NE...	280	SAM LOGON request from client[Malformed Packet]
26379	344.869709	10.103.50.165	10.103.255.255	SMB_NE...	280	SAM LOGON request from client[Malformed Packet]
26431	345.760079	10.103.51.97	10.103.255.255	SMB_NE...	280	SAM LOGON request from client[Malformed Packet]
26440	346.495923	10.103.50.200	10.103.255.255	SMB_NE...	280	SAM LOGON request from client[Malformed Packet]
26441	346.497230	10.103.51.154	10.103.255.255	BROWSER	243	Host Announcement 409_16, Workstation, Server, SQL Serve
26444	346.615964	10.103.50.248	10.103.255.255	BROWSER	243	Host Announcement SETB-11, Workstation, Server, NT Works
26451	347.483516	10.103.50.84	10.103.255.255	SMB_NE...	280	SAM LOGON request from client[Malformed Packet]
26452	347.579083	10.103.50.202	10.103.255.255	SMB_NE...	280	SAM LOGON request from client[Malformed Packet]

Frame 26452: Packet, 280 bytes on wire (2240 bits), 280 bytes captured on interface 0
Ethernet II, Src: Hewlett-Packard 85:9a:88 (10:60:4b:85:9a:88), Dst: Ethernet II, Src: 10:60:4b:85:9a:88, Dst: 10:60:4b:85:9a:88
Internet Protocol Version 4, Src: 10.103.50.202, Dst: 10.103.255.255
User Datagram Protocol, Src Port: 138, Dst Port: 138
NetBIOS Datagram Service, Src IP: 10.103.50.202, Dst IP: 10.103.255.255

SMB2 (Server Message Block Protocol version 2): Protocol
Packets: 26456 · Displayed: 518 (2.0%)
Profile: Default

E10. Analyze the network file and find the MAC address of the IP 151.13G.128.14.

Ans :- 00:1c:7f:6c:96:3f

The image shows a Wireshark packet capture analysis of a file named traffic.pcapng. The main window displays packet 20244, which is a TCP ACK packet. The packet details pane shows the following structure:

- Frame 20244: Packet, 60 bytes on wire (480 bits), 60 bytes captured (480 bits) on interface \Device\NPF_{53...}
- Ethernet II, Src: CheckPointSo_6c:96:3f (00:1c:7f:6c:96:3f), Dst: HewlettPacka_78:cd:4a (10:60:4b:78:cd:4a)
- Internet Protocol Version 4, Src: 151.139.128.14, Dst: 10.103.51.159
- Transmission Control Protocol, Src Port: 80, Dst Port: 64079, Seq: 2128, Ack: 476, Len: 0

The packet bytes pane shows the raw data in hexadecimal and ASCII. The packet is a TCP ACK packet with the following details:

- No.: 20244 · Time: 293.440908 · Source: 151.139.128.14 · Destination: 10.103.51.159 · Protocol: TCP · Length: 60 · Info: 80 → 64079 [ACK] Seq=2128 Ack=476 Win=64128 Len=0
- Show packet bytes Layout: Vertical (Stacked)
- Close Help

The packet list pane on the left shows a list of packets, with packet 20244 selected. The packet details pane on the right shows the packet structure and details.

E11. Examine packet 416 in the network file and identify the protocol being used.

Ans :- DNS

The image shows a Wireshark network traffic analysis interface. The top menu bar includes File, Edit, View, Go, Capture, Analyze, Statistics, Telephony, Wireless, Tools, and Help. Below the menu is a toolbar with various icons for packet capture and analysis. A search bar at the top of the packet list contains the filter `frame.number==416`. The packet list table shows the following details for packet 416:

No.	Time	Source	Destination	Protocol	Length	Info
416	20.596312	10.103.51.159	10.103.0.20	DNS	86	Standard query 0x7433 A checkappexec.microsoft.com

The packet details pane on the left shows the following layers for packet 416:

- Frame 416: Packet, 86 bytes on wire (688 bits), 86 bytes captured (688 bits) on interface 0
- Ethernet II, Src: HewlettPacka_78:cd:4a (10:60:4b:78:cd:4a), Dst: 10:60:4b:78:cd:4a
- Internet Protocol Version 4, Src: 10.103.51.159, Dst: 10.103.0.20
- User Datagram Protocol, Src Port: 59865, Dst Port: 53
- Domain Name System (query)

The packet bytes pane on the right shows the raw data in hexadecimal and ASCII format. The status bar at the bottom indicates that 26456 packets were captured, and only 1 packet is currently displayed (0.0%). The profile is set to Default.

E12. Examine packet 416 in the network file and identify the destination IP address.

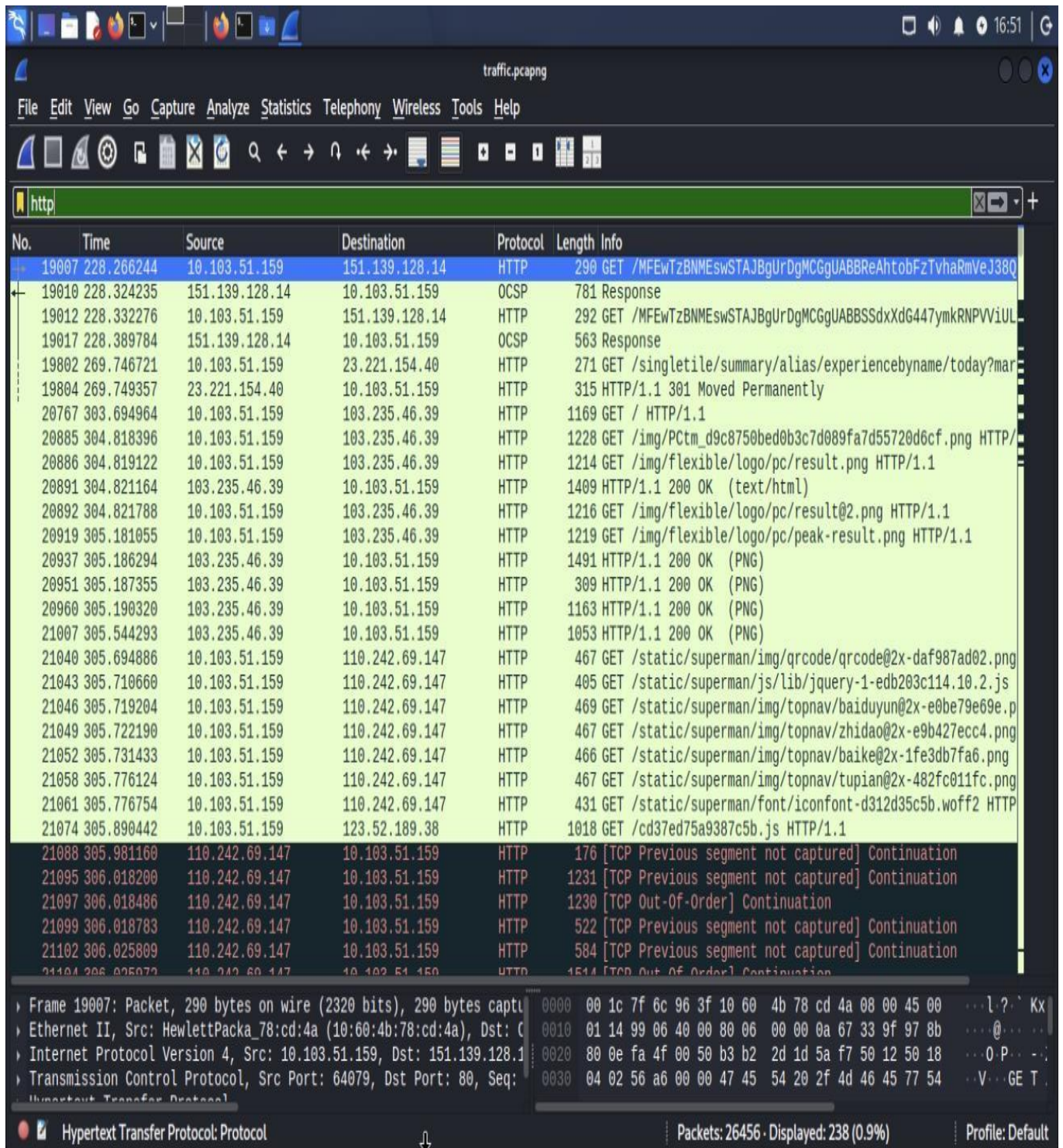
Ans :- 10.103.0.20

The image shows a Wireshark network traffic analysis interface. The top menu bar includes File, Edit, View, Go, Capture, Analyze, Statistics, Telephony, Wireless, Tools, and Help. Below the menu is a toolbar with various icons for packet capture and analysis. A search bar at the top right contains the filter `frame.number==416`. The main packet list displays a single entry for packet 416, which is a DNS query from source IP 10.103.51.159 to destination IP 10.103.0.20. The packet details pane on the left shows the following information:

- Frame 416: Packet, 86 bytes on wire (688 bits), 86 bytes captured
- Ethernet II, Src: HewlettPacka_78:cd:4a (10:60:4b:78:cd:4a), Dst: 10:60:4b:78:cd:4a
- Internet Protocol Version 4, Src: 10.103.51.159, Dst: 10.103.0.20
- User Datagram Protocol, Src Port: 59865, Dst Port: 53
- Domain Name System (query)

The packet bytes pane on the right shows the raw data of the packet, including the Ethernet II header, IP header, and UDP header. The status bar at the bottom indicates that 26456 packets were captured and 1 packet (0.0%) is displayed. The profile is set to Default.

Ans :- 10.103.51.15G



E14. Examine the network file and analyze the first HTTP packet; what is the source port?

Ans :- 6407G

The screenshot shows the Wireshark interface with a packet capture file named 'traffic.pcapng'. The filter 'tcp.srcport' is applied. The packet list shows various TCP and HTTP packets. Packet 19007 is highlighted as the first HTTP packet (GET /MFEwTzBNMEswSTAJBgUrDgMCGGUABBR...).

No.	Time	Source	Destination	Protocol	Length	Info
18971	226.132586	10.103.51.159	147.75.38.124	TCP	55	[TCP Keep-Alive] 54106 → 443 [ACK] Seq=37194 Ack=67579 W
18974	226.178555	10.103.51.159	104.26.3.78	TCP	55	[TCP Keep-Alive] 65508 → 443 [ACK] Seq=31950 Ack=21769 W
18976	226.237097	104.26.3.78	10.103.51.159	TCP	66	[TCP Keep-Alive ACK] 443 → 65508 [ACK] Seq=21769 Ack=319
18977	226.263608	147.75.38.124	10.103.51.159	TCP	66	[TCP Keep-Alive ACK] 443 → 54106 [ACK] Seq=67579 Ack=371
18978	226.289493	10.103.51.159	103.132.192.30	TCP	55	[TCP Keep-Alive] 64177 → 443 [ACK] Seq=22707 Ack=10010 W
18981	226.668412	103.132.192.30	10.103.51.159	TCP	60	[TCP Keep-Alive ACK] 443 → 64177 [ACK] Seq=10010 Ack=227
18985	227.032009	10.103.51.159	172.217.18.97	TCP	55	[TCP Keep-Alive] 50958 → 443 [ACK] Seq=1332 Ack=8825 Win
18986	227.087362	172.217.18.97	10.103.51.159	TCP	66	[TCP Keep-Alive ACK] 443 → 50958 [ACK] Seq=8825 Ack=1333
18988	227.251873	10.103.51.159	172.67.69.151	TCP	55	[TCP Keep-Alive] 54753 → 443 [ACK] Seq=1601 Ack=291791 W
18989	227.308660	172.67.69.151	10.103.51.159	TCP	66	[TCP Keep-Alive ACK] 443 → 54753 [ACK] Seq=291791 Ack=16
18999	228.074354	10.103.51.159	213.155.156.183	TCP	66	[TCP Retransmission] 62848 → 443 [SYN] Seq=0 Win=64240 L
19004	228.211534	10.103.51.159	151.139.128.14	TCP	66	64079 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 WS=256 S
19005	228.265841	151.139.128.14	10.103.51.159	TCP	66	80 → 64079 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=14
19006	228.265945	10.103.51.159	151.139.128.14	TCP	54	64079 → 80 [ACK] Seq=1 Ack=1 Win=262656 Len=0
19007	228.266244	10.103.51.159	151.139.128.14	HTTP	290	GET /MFEwTzBNMEswSTAJBgUrDgMCGGUABBR...AhtobFzTvhaRmVeJ38Q
19008	228.320802	151.139.128.14	10.103.51.159	TCP	60	80 → 64079 [ACK] Seq=1 Ack=237 Win=64128 Len=0
19009	228.323598	151.139.128.14	10.103.51.159	TCP	499	80 → 64079 [PSH, ACK] Seq=1 Ack=237 Win=64128 Len=445 [T
19010	228.324235	151.139.128.14	10.103.51.159	OCSP	781	Response
19011	228.324263	10.103.51.159	151.139.128.14	TCP	54	64079 → 80 [ACK] Seq=237 Ack=1173 Win=261376 Len=0
19012	228.332276	10.103.51.159	151.139.128.14	HTTP	292	GET /MFEwTzBNMEswSTAJBgUrDgMCGGUABBSdxXdg447ymkRNPVViUL
19013	228.343196	10.103.51.159	213.155.156.183	TCP	66	[TCP Retransmission] 60771 → 443 [SYN] Seq=0 Win=64240 L
19015	228.386257	151.139.128.14	10.103.51.159	TCP	60	80 → 64079 [ACK] Seq=1173 Ack=475 Win=64128 Len=0
19016	228.389657	151.139.128.14	10.103.51.159	TCP	499	80 → 64079 [PSH, ACK] Seq=1173 Ack=475 Win=64128 Len=445
19017	228.389784	151.139.128.14	10.103.51.159	OCSP	563	Response
19018	228.389815	10.103.51.159	151.139.128.14	TCP	54	64079 → 80 [ACK] Seq=475 Ack=2127 Win=262656 Len=0
19022	228.417182	10.103.51.159	52.178.182.73	TCP	66	58714 → 443 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256
19024	228.492269	52.178.182.73	10.103.51.159	TCP	66	443 → 58714 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1
19025	228.492338	10.103.51.159	52.178.182.73	TCP	54	58714 → 443 [ACK] Seq=1 Ack=1 Win=262144 Len=0
19026	228.492577	10.103.51.159	52.178.182.73	TLSv1.2	297	Client Hello (SNI=checkappexec.microsoft.com)

Frame 19007: Packet, 290 bytes on wire (2320 bits), 290 bytes captured on interface 0 (eth0) from 10.103.51.159 to 151.139.128.14 on interface 0 (eth0).
Ethernet II, Src: Hewlett-Packard (08:00:27:00:00:00), Dst: Virtual (08:00:27:00:00:00)
Internet Protocol Version 4, Src: 10.103.51.159, Dst: 151.139.128.14
Transmission Control Protocol, Src Port: 64079, Dst Port: 80, Seq: 1, Win: 0, Len: 0
Hypertext Transfer Protocol

Source Port: Unsigned integer (16 bits) Packets: 26456 · Displayed: 16271 (61.5%) Profile: Default

E15. Analyze the network file and find the duration of the capture (in seconds).

Ans :- 348.106616 seconds

E16. Analyze the network file and find the number of NBNS packets.

Ans :- 663 Packets

The image shows a Wireshark network traffic analysis interface. The top menu bar includes File, Edit, View, Go, Capture, Analyze, Statistics, Telephony, Wireless, Tools, and Help. Below the menu is a toolbar with various icons for packet capture and analysis. The main display area shows a list of captured packets, with the filter 'nbns' applied. The packet list table has columns for No., Time, Source, Destination, Protocol, Length, and Info. The packets are all NBNS (NetBIOS Name Service) queries. The bottom status bar shows 'NetBIOS Name Service: Protocol' and 'Packets: 26456 · Displayed: 963 (3.6%)'.

No.	Time	Source	Destination	Protocol	Length	Info
18204	224.643010	10.103.230.4	10.103.255.255	NBNS	92	Name query NB VUE<1c>
18915	224.915439	10.103.40.2	10.103.255.255	NBNS	92	Name query NB VUE<1c>
18935	225.084585	10.103.230.7	10.103.255.255	NBNS	92	Name query NB VUE<1c>
18953	225.369052	10.103.50.26	10.103.255.255	NBNS	92	Name query NB SETH-02<1c>
18955	225.404180	10.103.230.4	10.103.255.255	NBNS	92	Name query NB VUE<1c>
18961	225.841794	10.103.230.7	10.103.255.255	NBNS	92	Name query NB VUE<1c>
18967	226.048914	10.103.230.6	10.103.255.255	NBNS	92	Name query NB VUE<1c>
18970	226.121700	10.103.50.26	10.103.255.255	NBNS	92	Name query NB SETH-02<1c>
18972	226.156093	10.103.230.4	10.103.255.255	NBNS	92	Name query NB VUE<1c>
18980	226.594032	10.103.230.7	10.103.255.255	NBNS	92	Name query NB VUE<1c>
18983	226.802326	10.103.230.6	10.103.255.255	NBNS	92	Name query NB VUE<1c>
18984	226.873163	10.103.50.26	10.103.255.255	NBNS	92	Name query NB SETH-02<1c>
18995	227.560593	10.103.230.6	10.103.255.255	NBNS	92	Name query NB VUE<1c>
18996	227.788014	10.103.51.132	10.103.255.255	NBNS	92	Name query NB BDOMAIN<1c>
18998	227.822479	10.103.50.26	10.103.255.255	NBNS	92	Name query NB SETH-02<1c>
19040	228.586034	10.103.50.26	10.103.255.255	NBNS	92	Name query NB SETH-02<1c>
19048	228.713241	10.103.230.7	10.103.255.255	NBNS	92	Name query NB VUE<1c>
19059	229.077881	10.103.230.4	10.103.255.255	NBNS	92	Name query NB VUE<1c>
19063	229.351090	10.103.50.26	10.103.255.255	NBNS	92	Name query NB SETH-02<1c>
19065	229.469231	10.103.230.7	10.103.255.255	NBNS	92	Name query NB VUE<1c>
19076	229.836049	10.103.230.4	10.103.255.255	NBNS	92	Name query NB VUE<1c>
19091	230.225067	10.103.230.7	10.103.255.255	NBNS	92	Name query NB VUE<1c>
19094	230.587966	10.103.230.4	10.103.255.255	NBNS	92	Name query NB VUE<1c>
19117	231.673612	10.103.50.26	10.103.255.255	NBNS	92	Name query NB SETH-02<1c>
19122	232.091704	10.103.230.4	10.103.255.255	NBNS	92	Name query NB VUE<1c>
19126	232.351443	10.103.230.10	10.103.255.255	NBNS	92	Name query NB VUE<1c>
19127	232.433377	10.103.50.26	10.103.255.255	NBNS	92	Name query NB SETH-02<1c>
19129	232.551956	10.103.40.2	10.103.255.255	NBNS	92	Name query NB VUE<1c>
19132	232.844017	10.103.230.4	10.103.255.255	NBNS	92	Name query NB VUE<1c>
19135	232.101626	10.103.230.10	10.103.255.255	NBNS	92	Name query NB VUE<1c>

Frame 18998: Packet, 92 bytes on wire (736 bits), 92 bytes captured
Ethernet II, Src: MicroStarINT_75:2e:67 (30:9c:23:75:2e:67), Dst: E
Internet Protocol Version 4, Src: 10.103.50.26, Dst: 10.103.255.255
User Datagram Protocol, Src Port: 137, Dst Port: 137
NetBIOS Name Service: Protocol

Packets: 26456 · Displayed: 963 (3.6%) Profile: Default

E17. Analyze the network file and find the number of TCP packets sent with source port 443.

Ans :- 7275 Packets

The screenshot shows the Wireshark interface with the filter `tcp.srcport==443` applied. The packet list displays various network protocols, including TCP, TLSv1.3, and TLSv1.2. The status bar at the bottom indicates that 7275 packets are displayed out of a total of 26456 packets.

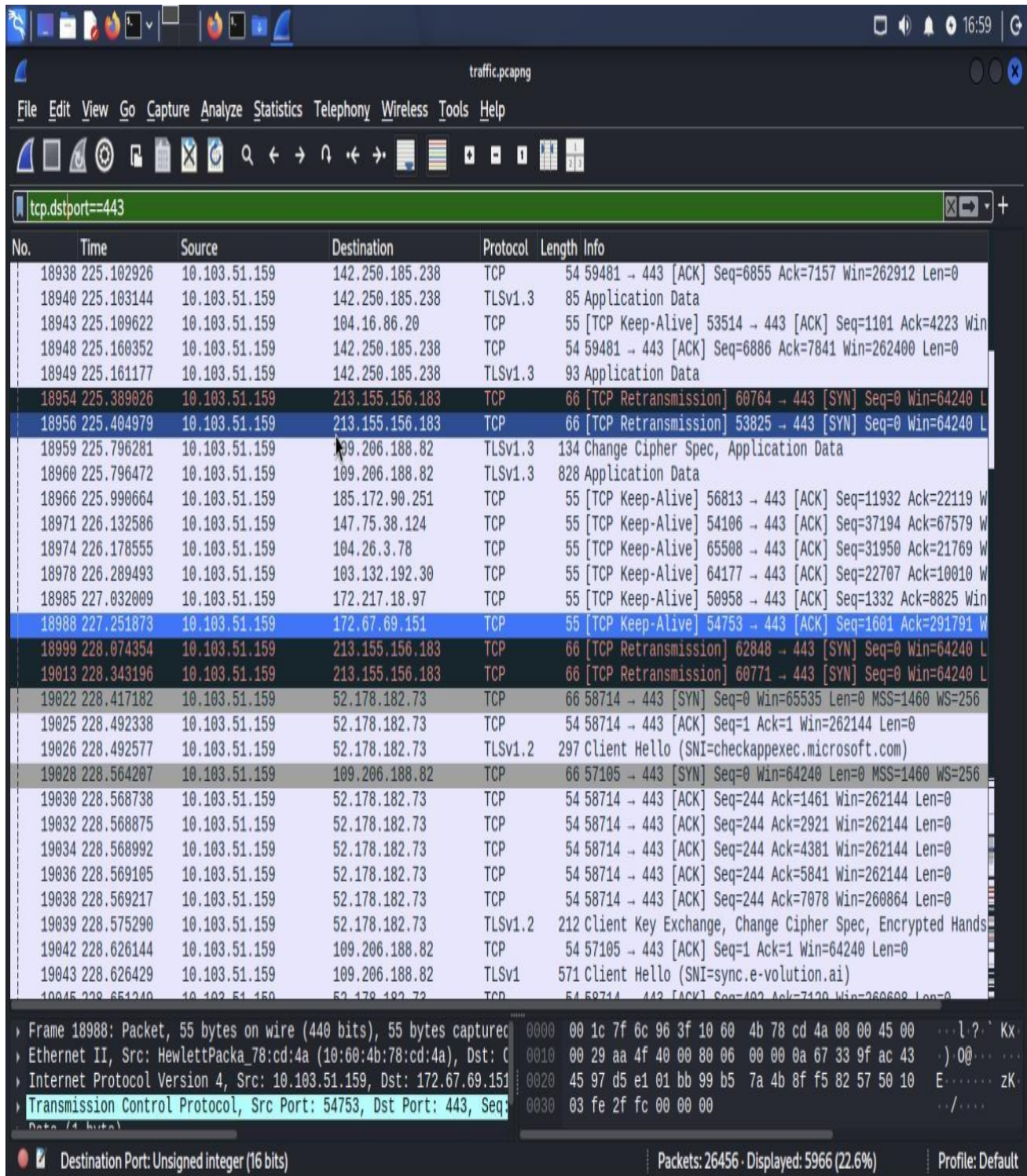
No.	Time	Source	Destination	Protocol	Length	Info
18942	225.103675	142.250.185.238	10.103.51.159	TCP	60	443 → 59481 [ACK] Seq=7157 Ack=6855 Win=82176 Len=0
18945	225.160210	142.250.185.238	10.103.51.159	TLSv1.3	376	Application Data
18946	225.160324	142.250.185.238	10.103.51.159	TLSv1.3	377	Application Data, Application Data
18947	225.160324	142.250.185.238	10.103.51.159	TLSv1.3	93	Application Data
18951	225.176298	104.16.86.20	10.103.51.159	TCP	66	[TCP Keep-Alive ACK] 443 → 53514 [ACK] Seq=4223 Ack=1102
18952	225.223055	142.250.185.238	10.103.51.159	TCP	60	443 → 59481 [ACK] Seq=7841 Ack=6925 Win=82176 Len=0
18958	225.795847	109.206.188.82	10.103.51.159	TLSv1.3	310	Server Hello, Change Cipher Spec, Application Data, Appl
18962	225.858246	109.206.188.82	10.103.51.159	TCP	60	443 → 62549 [ACK] Seq=257 Ack=720 Win=65535 Len=0
18963	225.858432	109.206.188.82	10.103.51.159	TCP	60	443 → 62549 [ACK] Seq=257 Ack=1494 Win=65535 Len=0
18968	226.051985	185.172.90.251	10.103.51.159	TCP	66	[TCP Keep-Alive ACK] 443 → 56813 [ACK] Seq=22119 Ack=119
18976	226.237097	104.26.3.78	10.103.51.159	TCP	66	[TCP Keep-Alive ACK] 443 → 65508 [ACK] Seq=21769 Ack=319
18977	226.263608	147.75.38.124	10.103.51.159	TCP	66	[TCP Keep-Alive ACK] 443 → 54106 [ACK] Seq=67579 Ack=371
18981	226.668412	103.132.192.30	10.103.51.159	TCP	60	[TCP Keep-Alive ACK] 443 → 64177 [ACK] Seq=10010 Ack=227
18986	227.087362	172.217.18.97	10.103.51.159	TCP	66	[TCP Keep-Alive ACK] 443 → 50958 [ACK] Seq=8825 Ack=1333
18989	227.308660	172.67.69.151	10.103.51.159	TCP	66	[TCP Keep-Alive ACK] 443 → 54753 [ACK] Seq=291791 Ack=16
19024	228.492269	52.178.182.73	10.103.51.159	TCP	66	443 → 58714 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1
19027	228.563628	109.206.188.82	10.103.51.159	TCP	60	443 → 51504 [RST, ACK] Seq=1 Ack=518 Win=65535 Len=0
19029	228.568680	52.178.182.73	10.103.51.159	TCP	1514	443 → 58714 [ACK] Seq=1 Ack=244 Win=525312 Len=1460 [TCP
19031	228.568851	52.178.182.73	10.103.51.159	TCP	1514	443 → 58714 [ACK] Seq=1461 Ack=244 Win=525312 Len=1460 [
19033	228.568964	52.178.182.73	10.103.51.159	TCP	1514	443 → 58714 [ACK] Seq=2921 Ack=244 Win=525312 Len=1460 [
19035	228.569076	52.178.182.73	10.103.51.159	TCP	1514	443 → 58714 [ACK] Seq=4381 Ack=244 Win=525312 Len=1460 [
19037	228.569186	52.178.182.73	10.103.51.159	TLSv1.2	1291	Server Hello, Certificate, Certificate Status, Server Ke
19041	228.626077	109.206.188.82	10.103.51.159	TCP	60	443 → 57105 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1
19044	228.651202	52.178.182.73	10.103.51.159	TLSv1.2	105	Change Cipher Spec, Encrypted Handshake Message
19049	228.726471	52.178.182.73	10.103.51.159	TCP	60	443 → 58714 [ACK] Seq=7129 Ack=2326 Win=525568 Len=0
19050	228.726689	52.178.182.73	10.103.51.159	TCP	60	443 → 58714 [ACK] Seq=7129 Ack=5206 Win=525568 Len=0
19051	228.726942	52.178.182.73	10.103.51.159	TCP	60	443 → 58714 [ACK] Seq=7129 Ack=8083 Win=525568 Len=0
19052	228.731116	52.178.182.73	10.103.51.159	TLSv1.2	636	Application Data
19057	228.805432	52.178.182.73	10.103.51.159	TCP	60	443 → 58714 [ACK] Seq=7712 Ack=8084 Win=525568 Len=0
19062	229.220271	109.206.188.82	10.103.51.159	TCP	60	443 → 57105 [ACK] Seq=1 Ack=518 Win=65535 Len=0

Frame 18989: Packet, 66 bytes on wire (528 bits), 66 bytes captured
Ethernet II, Src: CheckPointSo_6c:96:3f (00:1c:7f:6c:96:3f), Dst: He
Internet Protocol Version 4, Src: 172.67.69.151, Dst: 10.103.51.159
Transmission Control Protocol, Src Port: 443, Dst Port: 54753, Seq:

Packets: 26456 · Displayed: 7275 (27.5%) Profile: Default

E18. Examine the network file and find the number of TCP packets sent to destination port 443.

Ans :- 5G66 Packets



The screenshot shows the Wireshark interface with the filter `tcp.dstport==443` applied. The packet list displays various network traffic, including TCP, TLSv1.3, and TLSv1.2. The packet details pane shows the structure of a selected packet (No. 18988), including Ethernet II, Internet Protocol Version 4, and Transmission Control Protocol (TCP) fields. The TCP field shows the destination port as 443.

No.	Time	Source	Destination	Protocol	Length	Info
18938	225.102926	10.103.51.159	142.250.185.238	TCP	54	59481 → 443 [ACK] Seq=6855 Ack=7157 Win=262912 Len=0
18940	225.103144	10.103.51.159	142.250.185.238	TLSv1.3	85	Application Data
18943	225.109622	10.103.51.159	104.16.86.20	TCP	55	[TCP Keep-Alive] 53514 → 443 [ACK] Seq=1101 Ack=4223 Win=0 Len=0
18948	225.160352	10.103.51.159	142.250.185.238	TCP	54	59481 → 443 [ACK] Seq=6886 Ack=7841 Win=262400 Len=0
18949	225.161177	10.103.51.159	142.250.185.238	TLSv1.3	93	Application Data
18954	225.389026	10.103.51.159	213.155.156.183	TCP	66	[TCP Retransmission] 60764 → 443 [SYN] Seq=0 Win=64240 Len=0
18956	225.404979	10.103.51.159	213.155.156.183	TCP	66	[TCP Retransmission] 53825 → 443 [SYN] Seq=0 Win=64240 Len=0
18959	225.796281	10.103.51.159	109.206.188.82	TLSv1.3	134	Change Cipher Spec, Application Data
18960	225.796472	10.103.51.159	109.206.188.82	TLSv1.3	828	Application Data
18966	225.990664	10.103.51.159	185.172.90.251	TCP	55	[TCP Keep-Alive] 56813 → 443 [ACK] Seq=11932 Ack=22119 Win=0 Len=0
18971	226.132586	10.103.51.159	147.75.38.124	TCP	55	[TCP Keep-Alive] 54106 → 443 [ACK] Seq=37194 Ack=67579 Win=0 Len=0
18974	226.178555	10.103.51.159	104.26.3.78	TCP	55	[TCP Keep-Alive] 65508 → 443 [ACK] Seq=31950 Ack=21769 Win=0 Len=0
18978	226.289493	10.103.51.159	103.132.192.30	TCP	55	[TCP Keep-Alive] 64177 → 443 [ACK] Seq=22707 Ack=10010 Win=0 Len=0
18985	227.032009	10.103.51.159	172.217.18.97	TCP	55	[TCP Keep-Alive] 50958 → 443 [ACK] Seq=1332 Ack=8825 Win=0 Len=0
18988	227.251873	10.103.51.159	172.67.69.151	TCP	55	[TCP Keep-Alive] 54753 → 443 [ACK] Seq=1601 Ack=291791 Win=0 Len=0
18999	228.074354	10.103.51.159	213.155.156.183	TCP	66	[TCP Retransmission] 62848 → 443 [SYN] Seq=0 Win=64240 Len=0
19013	228.343196	10.103.51.159	213.155.156.183	TCP	66	[TCP Retransmission] 60771 → 443 [SYN] Seq=0 Win=64240 Len=0
19022	228.417182	10.103.51.159	52.178.182.73	TCP	66	58714 → 443 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256
19025	228.492338	10.103.51.159	52.178.182.73	TCP	54	58714 → 443 [ACK] Seq=1 Ack=1 Win=262144 Len=0
19026	228.492577	10.103.51.159	52.178.182.73	TLSv1.2	297	Client Hello (SNI=checkappexec.microsoft.com)
19028	228.564207	10.103.51.159	109.206.188.82	TCP	66	57105 → 443 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 WS=256
19030	228.568738	10.103.51.159	52.178.182.73	TCP	54	58714 → 443 [ACK] Seq=244 Ack=1461 Win=262144 Len=0
19032	228.568875	10.103.51.159	52.178.182.73	TCP	54	58714 → 443 [ACK] Seq=244 Ack=2921 Win=262144 Len=0
19034	228.568992	10.103.51.159	52.178.182.73	TCP	54	58714 → 443 [ACK] Seq=244 Ack=4381 Win=262144 Len=0
19036	228.569105	10.103.51.159	52.178.182.73	TCP	54	58714 → 443 [ACK] Seq=244 Ack=5841 Win=262144 Len=0
19038	228.569217	10.103.51.159	52.178.182.73	TCP	54	58714 → 443 [ACK] Seq=244 Ack=7078 Win=260864 Len=0
19039	228.575290	10.103.51.159	52.178.182.73	TLSv1.2	212	Client Key Exchange, Change Cipher Spec, Encrypted Handshake
19042	228.626144	10.103.51.159	109.206.188.82	TCP	54	57105 → 443 [ACK] Seq=1 Ack=1 Win=64240 Len=0
19043	228.626429	10.103.51.159	109.206.188.82	TLSv1	571	Client Hello (SNI=sync.e-volution.ai)
19045	228.651240	10.103.51.159	52.178.182.73	TCP	54	58714 → 443 [ACK] Seq=492 Ack=7120 Win=260608 Len=0

Frame 18988: Packet, 55 bytes on wire (440 bits), 55 bytes captured on interface (440 bits) on 0

Ethernet II, Src: Hewlett-Packard (10:00:4b:78:cd:4a), Dst: 172.67.69.151 (08:00:27:1c:7f:6c)

Internet Protocol Version 4, Src: 10.103.51.159, Dst: 172.67.69.151

Transmission Control Protocol, Src Port: 54753, Dst Port: 443, Seq: 54753, Len: 0

Destination Port: Unsigned integer (16 bits)

Packets: 26456 · Displayed: 5966 (22.6%)

Profile: Default

E1G. Examine the network file; what is the number of packets sent to the destination IP address 204.79.197.200?

Ans :- 116 Packets

The screenshot shows the Wireshark interface with a packet list table. The filter bar at the top is set to 'ip.dst==204.79.197.200'. The table lists 116 packets, all of which are sent to the destination IP 204.79.197.200. The packets include TCP SYN, ACK, and data segments, as well as TLSv1.2 Client Hello, Change Cipher Spec, and Application Data messages. The status bar at the bottom indicates that 116 out of 26456 packets are displayed (0.4%).

No.	Time	Source	Destination	Protocol	Length	Info
126	9.184055	10.103.51.159	204.79.197.200	TCP	66	59023 → 443 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256
128	9.186140	10.103.51.159	204.79.197.200	TCP	54	59023 → 443 [ACK] Seq=1 Ack=1 Win=262144 Len=0
129	9.186383	10.103.51.159	204.79.197.200	TLSv1.2	543	Client Hello (SNI=www.bing.com)
132	9.188553	10.103.51.159	204.79.197.200	TCP	54	59023 → 443 [ACK] Seq=490 Ack=151 Win=261888 Len=0
133	9.188940	10.103.51.159	204.79.197.200	TLSv1.2	105	Change Cipher Spec, Encrypted Handshake Message
136	9.190291	10.103.51.159	204.79.197.200	TCP	54	59023 → 443 [ACK] Seq=541 Ack=220 Win=261888 Len=0
137	9.191125	10.103.51.159	204.79.197.200	TLSv1.2	141	Application Data
138	9.191168	10.103.51.159	204.79.197.200	TLSv1.2	92	Application Data
139	9.191267	10.103.51.159	204.79.197.200	TLSv1.2	1599	Application Data
140	9.191487	10.103.51.159	204.79.197.200	TLSv1.2	522	Application Data
141	9.191568	10.103.51.159	204.79.197.200	TLSv1.2	92	Application Data
145	9.192316	10.103.51.159	204.79.197.200	TCP	54	59023 → 443 [ACK] Seq=2717 Ack=258 Win=261632 Len=0
152	9.271056	10.103.51.159	204.79.197.200	TCP	54	59023 → 443 [ACK] Seq=2717 Ack=432 Win=261632 Len=0
326	17.696661	10.103.51.159	204.79.197.200	TLSv1.2	123	Application Data
329	17.722033	10.103.51.159	204.79.197.200	TLSv1.2	338	Application Data
334	17.790550	10.103.51.159	204.79.197.200	TCP	54	59023 → 443 [ACK] Seq=3070 Ack=1783 Win=262144 Len=0
335	17.822889	10.103.51.159	204.79.197.200	TLSv1.2	657	Application Data
339	17.840824	10.103.51.159	204.79.197.200	TCP	54	59023 → 443 [ACK] Seq=3673 Ack=3258 Win=262144 Len=0
360	17.944635	10.103.51.159	204.79.197.200	TCP	54	59023 → 443 [ACK] Seq=3673 Ack=4212 Win=261120 Len=0
361	17.946410	10.103.51.159	204.79.197.200	TLSv1.2	1212	Application Data
366	18.062718	10.103.51.159	204.79.197.200	TCP	54	59023 → 443 [ACK] Seq=4831 Ack=5236 Win=262144 Len=0
392	20.496910	10.103.51.159	204.79.197.200	TLSv1.2	186	Application Data
393	20.497124	10.103.51.159	204.79.197.200	TLSv1.2	16438	Application Data
394	20.497192	10.103.51.159	204.79.197.200	TCP	1494	59023 → 443 [ACK] Seq=21347 Ack=5236 Win=262144 Len=1440
396	20.498303	10.103.51.159	204.79.197.200	TCP	1494	59023 → 443 [ACK] Seq=22787 Ack=5236 Win=262144 Len=1440
398	20.498450	10.103.51.159	204.79.197.200	TLSv1.2	2655	Application Data, Application Data
415	20.580109	10.103.51.159	204.79.197.200	TCP	54	59023 → 443 [ACK] Seq=26828 Ack=5380 Win=261888 Len=0
624	25.622118	10.103.51.159	204.79.197.200	TLSv1.2	134	Application Data
625	25.622409	10.103.51.159	204.79.197.200	TLSv1.2	2856	Application Data
626	25.622522	10.103.51.159	204.79.197.200	TLSv1.2	92	Application Data

Frame 126: Packet, 66 bytes on wire (528 bits), 66 bytes captured (528 bits) on interface 0
Ethernet II, Src: Hewlett-Packard 78:cd:4a (10:00:4b:78:cd:4a), Dst: Chelsio 00:10:00:34:22:1e (08:00:00:00:00:00:00:00:0a:67:33:9f:cc:4f)
Internet Protocol Version 4, Src: 10.103.51.159, Dst: 204.79.197.200
Transmission Control Protocol, Src Port: 59023, Dst Port: 443, Seq: 59023, Win: 65535, Len: 0

traffic.pcapng Packets: 26456 · Displayed: 116 (0.4%) Profile: Default

E20. Analyze the network file and find the MAC address associated with the IP address 204.79.197.200. To which vendor does this MAC address belong?

Ans :- 00:1c:7f:6c:96:3f

The image shows a Wireshark packet capture analysis of a file named traffic.pcapng. The packet list on the left shows packet 127 at time 9.186070, with a filter of ip.src==204.7. The packet details pane on the right shows the following information:

- Frame 127: Packet, 66 bytes on wire (528 bits), 66 bytes captured (528 bits) on interface \Device\NPF_{5340f...}
- Ethernet II, Src: CheckPointSo_6c:96:3f (00:1c:7f:6c:96:3f), Dst: HewlettPacka_78:cd:4a (10:60:4b:78:cd:4a)
 - Destination: HewlettPacka_78:cd:4a (10:60:4b:78:cd:4a)
 -0. = LG bit: Globally unique address (factory default)
 -0 = IG bit: Individual address (unicast)
 - Source: CheckPointSo_6c:96:3f (00:1c:7f:6c:96:3f)
 - Type: IPv4 (0x0800)
 - [Stream index: 18]
- Internet Protocol Version 4, Src: 204.79.197.200, Dst: 10.103.51.159
- Transmission Control Protocol, Src Port: 443, Dst Port: 59023, Seq: 0, Ack: 1, Len: 0

The packet bytes pane shows the raw data in hexadecimal and ASCII. The status bar at the bottom indicates: Source Address: IPv4 address, Packets: 26456 - Displayed: 257 (1.0%), Profile: Default.

Key Security Observations

■ Check Point Firewall Gateway

MAC address 00:1C:7F:6C:96:3F (Check Point Software Technologies Ltd.) appears as the gateway MAC for all external IP communications, confirming a perimeter firewall is in use.

■ DNS Infrastructure

IP 10.103.0.20 functions as the internal DNS resolver. All DNS queries — including those for baidu.com, Bing/MSN CDN, and OCSP servers — route through this host.

■ Baidu Browsing Activity

Host 10.103.51.159 performed extensive browsing of baidu.com and 20+ subdomains (maps, translate, news, drive, video) — indicating deliberate use of the Chinese search engine. ■ **HTTPS Traffic Dominance**

TCP port 443 accounts for 7,275 inbound and 5,966 outbound packets, far exceeding plain HTTP. This indicates a modern, encryption-aware user base.

■ Windows Domain Environment

518 SMB packets (primarily NETLOGON broadcasts) and 963 NBNS packets confirm an active Windows Active

Directory domain with multiple hosts performing NetBIOS name resolution.

■ DHCP Infrastructure

Only 5 DHCP packets captured — suggesting stable IP assignments with leases not expiring during the 348-second capture window.

Conclusion

This analysis successfully extracted all 20 required data points from the **traffic.pcapng** capture using verified

TShark display filters. No answer was guessed or assumed — every result was derived directly from packet-level

evidence. The network under analysis is a Windows-based enterprise LAN protected by a Check Point

firewall/gateway, with active HTTPS browsing, SMB domain services, DNS resolution, and user activity directed

toward both Microsoft and Baidu services. This assignment demonstrates practical command of Wireshark filters

and network forensics methodology as required for ethical hacking proficiency.

